Session 2: Shelves, cards and choices

Question: Which order lines are big?

Concepts

Concept	In daily life	In Excel	The limit
list	A row of bins on one shelf	One column of values	The first bin is number 0
len, sum, min, max	Count, total, smallest, largest	COUNT, SUM, MIN, MAX	Same idea, new spelling
dictionary	A product card: field name and value	One row with headers	Look up by name, not position
comparison	"Is this more than that?"	=A2>10	One = stores, two == ask
if, elif, else	A packing rule	=IF(...)	The 4-space indent is part of the rule

Code patterns

```
quantities = [6, 6, 8, 6, 6, 2, 6]
print(quantities[0], quantities[-1])
print(len(quantities), sum(quantities))

order_line = {"quantity": 6, "country": "United Kingdom"}
print(order_line["quantity"])

if quantity >= 100:
    print("Big")
elif quantity >= 12:
    print("Medium")
else:
    print("Small")
```

Three common errors

Error	What to do
IndexError	You asked for a bin that does not exist. The list has 7 items, so the last index is 6.
KeyError	The name is not on the card. Check the spelling of the key.
IndentationError / SyntaxError	A missing colon after if, or the lines under it are not indented.

Exercises and expected answers

Exercise	Expected
Predict: What is quantities[2]? And quantities[5]? Guess first.	Guess first, then run it to check
Fill in: The average quantity is the total divided by the count.	5.71
Fill in: The value of this line, from the card.	15.3
Fix it: This asks for a position that does not exist. Fix it so it prints the last quantity, 6.	The cell runs without an error
Fix it: This key is spelled wrong. Fix it so it prints 2.55.	The cell runs without an error
Predict: Set quantity to 12. Which line prints? Guess first.	Guess first, then run it to check
Write it: Print UK if the card's country is United Kingdom, else Outside UK.	UK
Write it: Add rows 8 to 10 to the list: 6, 6 and 6. Use quantities + [...]. Then print the new length and total.	10 and 58
Write it: A price rule: print Cheap if the price is below 3, else Not cheap. Use the card.	Cheap
Case: The warehouse packs a line in a big box when the quantity is 24 or more, else a small box. Make a card for the line with code 22752 (SET 7 BABUSHKA NESTING BOXES, 2 units at 7.65), and print which box it needs.	22752: small box
Extra: Change the card's country to Germany, then run your UK rule again.	Outside UK
Extra: How many different values are in the list? Use set(), which keeps each value once, and len.	3